1Z0-830 Java SE 21 模擬考題目與答案

補習班四次模擬考整合題庫，共 77 題，依單元排列；綠底為正確選項。題號與測驗程式的題號相同。

1. 基本語法與運算子（8 題）
2. 字串與文字區塊（3 題）
3. 類別、介面與物件導向（14 題）
4. Record（1 題）
5. Enum（3 題）
6. 樣式比對與 switch（4 題）
7. 例外處理（3 題）
8. 集合與陣列（7 題）
9. Stream API（10 題）
10. 並行與執行緒（8 題）
11. I/O、NIO 與序列化（6 題）
12. 日期時間、格式化與本地化（5 題）
13. 模組系統（5 題）

1. 基本語法與運算子

8 題

第 20 題

第1次模擬考第 20 題

Which two code fragments compile?

（選 2 個）

A.

```
class L5 {
    public void m() {
        var strVar = null;
    }
}
```

B.

```
class L2 {
    public void m(int x) {
        var x = 10;
    }
}
```

C.

```
class A {}
class B extends A {}
class L4 {
    public static void main(String[] args) {
        var x = new A();
        x = new B();
    }
}
```

D.

```
class L1 {
    var x = 100;
}
```

E.

```
class L3 {
    public static void main(String[] args) {
        var a = 10;
        a = "30";
    }
}
```

F.

```
class L6 {
    public static void main(String[] args) {
        var x = new ArrayList<>();
        x.add(10);
        x.add("30");
        System.out.println(x);
    }
}
```

答案：C, F

第 24 題

第1次模擬考第 24 題、第2次第 22 題、第3次第 22 題

Given:

```
public class Calculator {
    int calc(int i, String s) {
        return s.length()+i;
    }
    int calc(int i, int j) {
        return i*j;
    }
    float calc(String s) {
        return s.length()/2f;
    }
    float calc(int i) {
        return i/2f;
    }
    public static void main(String[] args) {
        Calculator cal = new Calculator();
        var a = "ACME";
        var b = cal.calc(a);
        // Line n1
        System.out.println(c+" "+d+" "+e);
    }
}
```

Which code fragment inserted at line marked "Line n1" would result in the following output: `6 3.0 36`?

A.

```
var c = cal.calc(Math.round(b));
var d = cal.calc(Math.round(c),a);
var e = cal.calc(d, d);
```

B.

```
var c = cal.calc(Math.round(b));
var d = cal.calc(Math.round(c),a);
var e = cal.calc(d,a);
```

C.

```
var c = cal.calc(Math.round(b),a);
var d = cal.calc(Math.round(c));
var e = cal.calc(c, c);
```

D.

```
var c = cal.calc(Math.round(b), Math.round(b));
var d = cal.calc(c,c);
var e = cal.calc(Math.round(b),a);
```

答案：C

第 25 題

第1次模擬考第 25 題、第3次第 21 題

Given:

```
public class Question {
    public void answer(int a, int b) {
        System.out.print(" A");
```

```
    }
    public void answer(int a, float b) {
        System.out.print(" B");
    }
    public void answer(float a, float b) {
        System.out.print(" C");
    }
    public void answer(double... a) {
        System.out.print(" D");
    }
    public static void main(String[] args) {
        Question q = new Question();
        q.answer(10,15.25);
        q.answer(10, 24);
        q.answer(10.25,10.25);
    }
}
```

What is the result?

A. `B A C`

B. `B A D`

C. `D A D`

D. `D D D`

答案：C

第 30 題 第1次模擬考第 30 題、第3次第 5 題

Given the code fragment:

```
int x = Integer.valueOf("a", 16)-9;
String y = "128";
byte z = Boolean.valueOf(String.valueOf(x)) ? Integer.valueOf(y).byteValue() : (byte)x;
System.out.println(z);
```

What is the result?

A. `0`

B. `128`

C. `-128`

D. Exception is thrown at run time

E. `1`

F. `-1`

答案：E

第 31 題 第1次模擬考第 31 題、第2次第 1 題

Given the code fragment:

```
int i = 15;
// Line n1
System.out.println(j);
```

Which statement inserted at "Line n1" enables you to print `2.0`?

A. `float j = i % 3 + 2;`

B.
```
float j = i / 3 + 2;
```

C.
```
float j = i / (3 + 2);
```

D.
```
float j = i % (3 + 2);
```

答案：A

第 51 題

第2次模擬考第 2 題、第3次第 4 題

Given the code fragment:

```
int a = 2;
int b = ~a;
int c = a^b;
boolean d = a < b & a > c++;
System.out.println(d + " " + c);
boolean e = a > b && a > c++;
System.out.println(e + " " + c);
```

What is the result?

A.
```
false 0
true 1
```

B.
```
true 1
false 2
```

C.
```
false 1
false 2
```

D.
```
false 0
true 2
```

E.
```
false 1
true 2
```

答案：A

第 66 題

第2次模擬考第 48 題、第3次第 45 題

Given the code fragment:

```
int c = 0;
x:
for (int a = 3; a > 0; a--) {
    for (int b = 0; b < 3; b++, c--) {
        if (b == 1) continue x; else if (a == 1) break; c += a+b;
    }
}
System.out.println(c);
```

What is the result?

A.
```
5
```

B.
```
3
```

C.
```
4
```

D.
```
2
```

答案：B

第 67 題 第2次模擬考第 50 題、第3次第 43 題

Given the code fragment:

```
int a = 0, b = 1, c = 2;
if (Math.round(b/c) < b/c & b != b/++a) {
    c += a+++b;
} else {
    c -= --b+a;
}
System.out.println(c);
```

What is the result?

A. `2`

B. `1`

C. `-1`

D. A runtime exception is thrown

E. `0`

答案：B

2. 字串與文字區塊

3 題

第 42 題 第1次模擬考第 42 題、第3次第 23 題

Given the code fragment:

```
// Line n1
String s = "-%s-";
System.out.printf(s, x.charAt(6));
```

Which statement inserted at "Line n1" enables you to print `-S-`?

A.
```
String x = """
Java
SE
""";
```

B.
```
String x = """
Java SE""";
```

C.
```
String x = """Java SE""";
```

D.
```
String x = """
Java
 SE
""";
```

答案：D

第 43 題 第1次模擬考第 43 題、第2次第 42 題、第3次第 24 題

Given the code fragment:

```
String a = "Hello! Java";
System.out.print(a.indexOf("Java"));
a.replace("Hello!", "Welcome!");
System.out.print(a.indexOf("Java"));
StringBuilder b = new StringBuilder(a);
System.out.print(b.indexOf("Java"));
```

What is the result?

A. `888`

B. `7107`

C. `81111`

D. `777`

E. `71010`

F. `8109`

G. `799`

答案：D

第 63 題 第2次模擬考第 43 題

Given the code fragment:

```
String myStr = "Hello Java";
String myTextBlk1 = """
                    Hello Java""";
String myTextBlk2 = """
                    Hello Java
                    """;
System.out.print(myStr.equals(myTextBlk1)+":");
System.out.print(myStr.equals(myTextBlk2)+":");
System.out.print(myTextBlk1.equals(myTextBlk2)+":");
System.out.println(myTextBlk1.intern() == myTextBlk2.intern());
```

What is the result?

A. `true:true:false:false`

B. `true:false:true:false`

C. `true:false:false:false`

D. `true:false:true:true`

答案：C

3. 類別、介面與物件導向

14 題

第 12 題 第1次模擬考第 12 題、第2次第 21 題、第3次第 40 題

Given:

```
public class Symbols {
    private char value = '?';
    public Symbols(char value) {
        this.value = value;
    }
    public Symbols() {
        this('!');
    }
    static { System.out.print('/'); }
    { System.out.print(this.value); }
    public String toString() {
        return String.valueOf(this.value);
    }
    public static void main(String[] args) {
```

```
        Symbols s1 = new Symbols();
        Symbols s2 = new Symbols('-');
        String symbols = s1.toString()+s2.toString();
        System.out.print(symbols);
    }
}
```

What is the result?

A. `!?-?!`

B. `/?!-`

C. `/??!-`

D. `/!!-`

E. `!??-!`

F. `!/?!-`

答案：C

第 21 題 第1次模擬考第 21 題、第2次第 37 題

Which example defines the `Book` class to provide immutable `Book` objects?

A.
```
public final class Book {
    private static int maxId;
    private final int id;
    private final String title;
    public Book () {
        this.id = ++maxId;
    }
    public Book (String title) {
        this.title = title;
    }
    public String getTitle() {
        return title;
    }
}
```

B.
```
public final class Book {
    private static int maxId;
    private final int id;
    private final String title;
    public Book () {
        this("");
        id = ++maxId;
    }
    public Book (String title) {
        this.title = title;
    }
    public String getTitle() {
        return title;
    }
}
```

C.

```
public final class Book {
    private static int maxId;
    private final int id;
    private final String title;
    public Book () {
        id = ++maxId;
    }
    public Book (String title) {
        this.id = ++maxId;
        this.title = title;
    }
    public String getTitle() {
        return title;
    }
}
```

D.

```
public final class Book {
    private static int maxId;
    private final int id;
    private final String title;
    public Book () {
        this("");
    }
    public Book (String title) {
        this.id = ++maxId;
        this.title = title;
    }
    public String getTitle() {
        return title;
    }
}
```

E.

```
public final class Book {
    private static int maxId;
    private final int id = ++maxId;
    private final String title;
    public Book () {
    }
    public Book (String title) {
        this();
        this.title = title;
    }
    public String getTitle() {
        return title;
    }
}
```

答案：D

第 34 題

第1次模擬考第 34 題、第2次第 9 題、第3次第 25 題

Given:

```
public final class Music extends Art {
    private String metre;
    public Music(String title, String metre) {
        super(title);
        this.metre = metre;
    }
    public String metre() {
        return metre;
    }
}
```

```
public non-sealed class Dance extends Art {
    private String metre;
    public Dance(String title, String metre) {
        super(title);
        this.metre = metre;
    }
    public String metre() {
        return metre;
    }
}

public class Stage {
    public static void main(String[] args) {
        Art a1 = new Music("Waltz","3/4");
        Art a2 = new Dance("Waltz","3/4");
        Art a3 = new Dance("Waltz","3/4");
        System.out.print(a1.equals(a2)+" ");
        System.out.print(a2.equals(a3)+" ");
        System.out.print(a3.equals(null)+" ");
    }
}
```

Which definition of the class `Art` allows to produce the `false true false` output?

A.

```
public sealed class Art permits Music, Dance {
    private String title;
    public Art(String title){
        this.title = title;
    }
    public boolean equals(Object obj) {
        return switch (obj) {
            case Dance dance -> dance.metre().equals(this.metre());
            case Music music -> music.metre().equals(this.metre());
            case null -> false;
            default -> false;
        };
    }
}
```

B.

```
public sealed class Art permits Music, Dance {
    private String metre;
    public Art(String metre){
        this.metre = metre;
    }
    public String metre() {
        return metre;
    }
    public boolean equals(Object obj) {
        if (obj instanceof Art art) {
            return art.metre().equals(this.metre());
        }
        return false;
    }
}
```

C.

```
public sealed class Art permits Music, Dance {
    private String title;
    public Art(String title){
        this.title = title;
    }
    public boolean equals(Object obj) {
        return obj != null && obj.getClass().equals(this.getClass())
            && (((Dance)obj).metre().equals(metre())
            || (((Music)obj).metre().equals(metre())));
    }
}
```

D.

```
public abstract sealed class Art permits Music, Dance {
    private String title;
    public Art(String title){
        this.title = title;
    }
    public abstract String metre();
    public boolean equals(Object obj) {
        return obj != null && obj.getClass().equals(this.getClass())
            && ((Art)obj).metre().equals(this.metre());
    }
}
```

E.

```
public abstract sealed class Art permits Music {
    private String title;
    public Art(String title){
        this.title = title;
    }
    public abstract String metre();
    public boolean equals(Object obj) {
        if (obj instanceof Art art) {
            return art.metre().equals(this.metre());
        }
        return false;
    }
}
```

答案：D

第 40 題

第1次模擬考第 40 題、第2次第 5 題

Given:

```
interface IFace {
    public void m1();
    public default void m2() {
        System.out.println("m2");
    }
    public static void m3() {
        System.out.println("m3");
    }
    private void m4() {
        System.out.println("m4");
    }
}

class MyC implements IFace {
    public void m1() {
        System.out.println("Hello");
    }
}
```

Which two method invocations execute?

（選 2 個）

A.
```
new MyC().m2();
```

B.
```
IFace.m2();
```

C.
```
IFace myClassObj = new MyC();
myClassObj.m4();
```

D.
```
IFace myClassObj = new MyC();
myClassObj.m3();
```

E.
```
IFace.m3();
```

F.
```
IFace.m4();
```

答案：A, E

第 41 題

第1次模擬考第 41 題、第2次第 6 題、第3次第 36 題

Given:

```
public interface Magic {
    int MAX_POWER = 10;
    public default boolean applyPower(int power) {
        return getRemainingPower(power) >= 0;
    }
    static int getRemainingPower(int power) {
        return MAX_POWER-power;
    }
    int cast();
    String toString();
}

public class Spell implements Magic {
    // Line n1
    private int power;
    public Spell(int power) {
        this.power = power;
    }
    public static void main(String[] args) {
        try {
            Spell spell = new Spell(6);
            int power = spell.cast();
            System.out.println("Power="+spell.power+" Overhead="+power);
        } catch (Exception e) {
            System.out.println(e.getMessage());
        }
    }
}
```

Which code fragment correctly implements the `Magic` interface when inserted into the `Spell` record at line marked as "Line n1"?

A.
```
@Override
public int cast() {
  if (applyPower(power)) {
    return getRemainingPower(power);
  } else {
    throw new RuntimeException("Not enough power");
  }
}
```

B.

```
@Override
public int cast() {
  if (applyPower(power)) {
    return getRemainingPower(power);
  } else {
    throw new RuntimeException("Not enough power");
  }
}
@Override
public String toString() {
  return "Power="+this.power+" Overhead="+getRemainingPower(power);
}
```

C.

```
@Override
public int cast() throws RuntimeException {
  if (applyPower(power)) {
    return Magic.getRemainingPower(power);
  } else {
    throw new RuntimeException("Not enough power");
  }
}
```

D.

```
@Override
public int cast() throws Exception {
  if (applyPower(power)) {
    return Magic.getRemainingPower(power);
  } else {
    throw new Exception("Not enough power");
  }
}
```

E.

```
@Override
public int cast() throws Exception {
  if (applyPower(power)) {
    return Magic.getRemainingPower(power);
  } else {
    throw new Exception("Not enough power");
  }
}
public String toString() {
  return "Power="+this.power+" Overhead="+Magic.getRemainingPower(power);
}
```

答案：C

第 44 題

第1次模擬考第 44 題、第2次第 41 題

Given:

```
public class App {
    private App app;
    public App() {
        this.app = null;
    }
    public App(App app) {
        this.app = app;
    }
    public String play() {
        return "playing";
    }
    public static void main(String[] args) {
        App a1 = new App();
        App a2 = new App(a1);
```

```
        App a3 = new App();
        a1 = null;
        System.out.println(a2.play());
    }
}
```

Which statement is true while the program prints the word "playing"?

A. All of the App instances are eligible for garbage collection

B. None of the App instances are eligible for garbage collection

C. Only one instance of the App is eligible for garbage collection

D. Two instances of the App are eligible for garbage collection

答案：B

第 45 題

第1次模擬考第 45 題、第2次第 40 題、第3次第 2 題

Which program prints `Bibbidi-Bobbidi-Boo` and `Abracadabra`?

A.

```
public class Wizard {
    record Spell(String cast) { }
    public static void main(String[] args) {
        Spell[] spells = {new Spell("Bibbidi-Bobbidi-Boo"),
                          new Spell("Abracadabra")};
        class Duel {
            void fight() {
                for (Spell spell: spells) {
                    System.out.println(spell.cast());
                }
            }
        }
        Duel duel = new Duel();
        duel.fight();
    }
}
```

B.

```
public class Wizard {
    record Spell(String cast) { }
    public static void main(String[] args) {
        Spell[] spells = null;
        class Duel {
            void fight() {
                System.out.println(spells[0].cast());
                System.out.println(spells[1].cast());
            }
        }
        spells = new Spell[2];
        spells[0] = new Spell("Bibbidi-Bobbidi-Boo");
        spells[1] = new Spell("Abracadabra");
        Duel duel = new Duel();
        duel.fight();
    }
}
```

C.

```
public class Wizard {
    class Spell {
        public String cast;
        public Spell (String cast) {
            this.cast = cast;
        }
    }
    private static Spell[] spells;
    public static void main(String[] args) {
        class Duel {
            void fight() {
                for (Spell spell: spells) {
                    System.out.println(spell.cast);
                }
            }
        }
        spells = new Spell[]{new Spell("Bibbidi-Bobbidi-Boo"),
                             new Spell("Abracadabra")};
        Duel duel = new Duel();
        duel.fight();
    }
}
```

D.

```
public class Wizard {
    class Spell {
        String cast;
        Spell (String cast) {
            this.cast = cast;
        }
    }
    private static Spell[] spells = {new Spell("Bibbidi-Bobbidi-Boo"),
                                     new Spell("Abracadabra")};
    public static void main(String[] args) {
        class Duel {
            void fight() {
                for (Spell spell: spells) {
                    System.out.println(spell.cast);
                }
            }
        }
        new Duel().fight();
    }
}
```

答案：A

第 56 題

第2次模擬考第 23 題

Given:

```
public class Test {
    interface Animal {                  // 這幾行照片沒拍到，依畫面殘影重建
        default String play() {
            return "jump";
        }
    }
    static class Dog implements Animal {
        public String play() {
            return "fetch";
        }
    }
    static class Cat implements Animal {
        public String play() {
```

```
            return "scratch";
        }
    }
    private static void play(Animal a) {
        System.out.println(a.play());
    }
    private static void play(Dog d) {
        System.out.println(d.play());
    }
    public static void main(String[] args) {
        Animal a1 = new Dog();
        Dog a2 = new Dog();
        Animal a3 = new Cat();
        play(a1);
        play(a2);
        play(a3);
    }
}
```

What is the result?

A.
```
jump
fetch
scratch
```

B.
```
jump
fetch
jump
```

C.
```
fetch
fetch
jump
```

D.
```
fetch
fetch
scratch
```

E. Compilation error

答案：D

第 62 題 第2次模擬考第 36 題、第3次第 34 題

Which code fragment prints the message `Hi!`?

A.
```
public class App {
    String txt = "Hi";
    static String message;
    public static void main(String... args) {
        App a = new App();
        var txt = "Hello";
        message = txt;
        a.greet('.');
        System.out.println(message);
    }
    String greet(char sign) {
        sign = '!';
        return message+sign;
    }
}
```

B.

```
public class App {
    String txt = "Hi";
    static String message;
    public static void main(String... args) {
        App a = new App();
        var message = a.txt;
        a.greet('.');
        System.out.println(message);
    }
    String greet(char sign) {
        sign = '!';
        return message+sign;
    }
}
```

C.

```
public class App {
    String txt = "Hi";
    static String message;
    public static void main(String... args) {
        App a = new App();
        a.txt = "Hello";
        var message = a.greet('.');
        System.out.println(message);
    }
    String greet(char sign) {
        sign = '!';
        return message+sign;
    }
}
```

D.

```
public class App {
    String txt = "Hi";
    static String message;
    public static void main(String... args) {
        App a = new App();
        message = a.txt;
        message = a.greet('.');
        System.out.println(message);
    }
    String greet(char sign) {
        sign = '!';
        return message+sign;
    }
}
```

E.

```
public class App {
    String txt = "Hi";
    static String message;
    public static void main(String... args) {
        App a = new App();
        message = a.txt;
        message = a.greet('.');
        System.out.println(message);
    }
    String greet(char sign) {
        var sign = '!';
        return message+sign;
    }
}
```

答案：D

第 69 題 第3次模擬考第 3 題

Given:

```
class OClass {
    private int var1 = 100;
    public int var2 = 200;
    class IClass {
        int var3 = 300;
        public int var4 = 400;
        int m2(){
            return var2;
        }
    }
    public int m1(){
        return var1;
    }
}

// and the code fragment:

public class App {
    public static void main(String[] args) {
        OClass.IClass inner = new OClass().new IClass();
    }
}
```

Which three members can be accessed by the `inner` reference variable?

（選 3 個）

A. `var4`

B. `m2`

C. `m1`

D. `var2`

E. `var3`

F. `var1`

答案：A, B, E

第 71 題 第3次模擬考第 35 題

Given:

```
public class App {
    public String x = "01";
    public static void main(String[] args) {
        String x = "10";
        App t = new App();
        t.myMethod(x);
        System.out.print(x);
    }
    public void myMethod(String x) {
        x += "1";
        System.out.print(x);
        System.out.print(this.x);
    }
}
```

What is the result?

A.
```
10110110
```

B.
```
1010110
```

C.
```
0110110
```

D.
```
1001110
```

E.
```
0111010
```

答案：B

第 72 題 第3次模擬考第 37 題

Which two interfaces are considered to be functional interfaces?

（選 2 個）

A.
```
@FunctionalInterface
interface Magic {
  int MAX_POWER = 10;
  public default boolean applyPower(int power) {
    return getRemainingPower(power) >= 0;
  }
  static int getRemainingPower(int power) {
    return MAX_POWER-power;
  }
}
```

B.
```
@FunctionalInterface
interface Magic {
  int MAX_POWER = 10;
  public default boolean applyPower(int power) {
    return getRemainingPower(power) >= 0;
  }
  int getRemainingPower(int power);
  int cast();
}
```

C.
```
interface Magic {
  int MAX_POWER = 10;
  public default void applyPower(int power) {
    return MAX_POWER = power;
  }
}
```

D.
```
interface Magic {
  int MAX_POWER = 10;
  public default boolean applyPower(int power) {
    return getRemainingPower(power) >= 0;
  }
  private int getRemainingPower(int power) {
    return MAX_POWER-power;
  }
  int cast();
}
```

E.
```
@FunctionalInterface
interface Magic {
  int MAX_POWER = 10;
  public default boolean applyPower(int power) {
    return getRemainingPower(power) >= 0;
  }
  static int getRemainingPower(int power) {
    return MAX_POWER-power;
  }
  int cast();
}
```

F.
```
@FunctionalInterface
interface Magic {
  private int MAX_POWER = 10;
  public default boolean applyPower(int power) {
    return getRemainingPower(power) >= 0;
  }
  int getRemainingPower(int power);
}
```

答案：D, E

第 76 題

第4次模擬考第 30 題

Which example correctly defines and implements an interface?

A.
```
public sealed interface Book permits Story, Art {
    default String title() { return "Winnie the Pooh" ; }
}
class Story implements Book {}
class Art implements Book {}
```

B.
```
non-sealed interface Book permits Story {
    String getTitle() { return "Winnie the Pooh" ; }
}
class Story implements Book {}
```

C.
```
interface Book {
    public default int getCounter() { return 10; }
    public static void displayTitle();
    public void shareFile();
}
class Story implements Book {}
```

D.
```
interface Book {
    public default int getCounter() { return 10; }
    public final void title() {
        System.out.println("Winnie the Pooh");
    }
}
class Story implements Book {}
```

E.
```
interface Book {
    public static final String title = "Winnie the Pooh";
    private void readStory() {
        System.out.println(title);
    }
}
class Story implements Book {}
```

F.

```
interface Book {
    public default int getCounter() { return 10; }
    public void title();
}
class Story implements Book {
    String title() { return "Winnie the Pooh" ; }
}
```

答案：E

第 77 題 第4次模擬考第 43 題

Given:

```
public abstract class Animal {
    public static enum Type {
        DOG, CAT, FISH;
    }
    private Type type;
    public Animal(Type type) {
        this.type = type;
    }
    public void setType(Type type) {
        this.type = type;
    }
    public abstract String name();
    public String toString() {
        return name()+" "+type;
    }
    public static void main(String[] args) {
        Animal a = new Pet("Fido", Animal.Type.DOG);
        System.out.println(a);
    }
}

public interface Playable {
    default String play() {
        return "none";
    }
}
```

Which two alternative definitions of the class `Pet` enable the program to print `Fetch Fido DOG`?

（選 2 個）

A.

```
public class Pet extends Animal implements Playable {
    private String name;
    public Pet(String name, Type type) {
        super(type);
        this.name = name;
    }
    public String name() {
        return play()+" "+name;
    }
    public String toString() {
        return "Fetch "+name+" "+super.type;
    }
}
```

B.

```
public class Pet extends Animal implements Playable {
    private String name;
    public Pet(String name, Type type) {
        super(type);
        this.name = name;
    }
    public String name() {
        return play()+" "+name;
    }
    public String play() {
        return "";
    }
    public String toString() {
        return "Fetch"+super.toString();
    }
}
```

C.

```
public class Pet extends Animal implements Playable {
    private String name;
    public Pet(String name, Type type) {
        this.name = name;
        super.setType(type);
    }
    public String name() {
        return play()+" "+name;
    }
    public String play() {
        return "Fetch";
    }
}
```

D.

```
public class Pet extends Animal implements Playable {
    private String name;
    public Pet(String name, Type type) {
        super(type);
        this.name = name;
    }
    public String name() {
        return play()+" "+name;
    }
    String play() {
        return "Fetch";
    }
}
```

E.

```
public class Pet extends Animal implements Playable {
    private String name;
    public Pet(String name, Type type) {
        super(type);
        this.name = name;
    }
    public String name() {
        return play()+" "+name;
    }
    public String play() {
        return "Fetch";
    }
}
```

答案：B, E

4. Record

1 題

第 13 題

第1次模擬考第 13 題、第2次第 20 題、第3次第 41 題

Which two record definitions are valid?

（選 2 個）

A.

```
record Item(String name) {
    private static int maxId;
    private int id;
    {
        id = ++maxId;
    }
    public Item {
        if (name.length() < 3) {
            throw new IllegalArgumentException();
        }
    }
}
```

B.

```
record Item(String name){
    private static int maxId;
    private int id;
    public Item(String name){
        this.id = ++maxId;
        if (name.length() < 3){
            throw new IllegalArgumentException();
        }else{
            this.name = name;
        }
    }
}
```

C.

```
record Item(int id, String name){
    private static int maxId;
    public Item(int id, String name){
        this.id = ++maxId;
        if (name.length() < 3){
            throw new IllegalArgumentException();
        }else{
            this.name = name;
        }
    }
}
```

D.

```
record Item(int id, String name) {
    private static int maxId;
    {
        id = ++maxId;
    }
    public Item {
        if (name.length() < 3) {
            throw new IllegalArgumentException();
        }
    }
}
```

E.
```
record Item(int id, String name){
    private static int maxId;
    public Item {
        id = ++maxId;
        if (name.length() < 3){
            throw new IllegalArgumentException();
        }
    }
}
```

F.
```
record Item(int id, String name){
    private static int maxId;
    public Item {
        this.id = ++maxId;
        if (name.length() < 3){
            throw new IllegalArgumentException();
        }else{
            this.name = name;
        }
    }
}
```

答案：C, E

5. Enum

3 題

第 26 題

第1次模擬考第 26 題、第2次第 30 題、第3次第 15 題

Given:

```
public class Weather {
    public enum Forecast {
        SUNNY, CLOUDY, RAINY;
        public String toString() {
            return this.name().toLowerCase();
        }
        static String getForecast(String name) {
            try {
                Forecast f = Forecast.valueOf(name);
                return (f.ordinal() % 2 != 0)
                    ? f.name().toLowerCase() : f.name().toUpperCase();
            }catch(Exception e){
                return "SNOWY";
            }
        }
    }
    public static void main(String[] args) {
        for (int i = 0; i < Forecast.values().length; i++){
            String name = Forecast.values()[i].toString();
            System.out.print(Forecast.getForecast(name));
        }
    }
}
```

What is the result?

A. `SUNNYCLOUDYRAINY`

B. `SNOWYSNOWYSNOWY`

C. `SUNNYcouldyRAINY`

D. `snowysnowysnowy`

E.
```
sunnyCLOUDYrainy
```

F.
```
sunnycouldyrainy
```

答案：B

第 27 題

第1次模擬考第 27 題、第2次第 29 題、第3次第 14 題

Given the `Test` class:

```
public class Test {
    public static void main(String[] value) {
        List<String> list = List.of("Hello","and","WELCOME");
        String result = list.stream()
            .collect(Collectors.mapping(v->Transformer.transform(v),
                    Collectors.joining(" ")));
        System.out.println(result);
    }
}
```

Which implementation of the `Transformer` enum produces the result `"Hello AND welcome"`?

A.
```
import java.util.function.UnaryOperator;
public enum Transformer {
    UPPER((String value) -> value.toUpperCase()),
    LOWER((String value) -> value.toLowerCase());
    private final UnaryOperator<String> operator;
    public Transformer(UnaryOperator<String> operator) {
        this.operator = operator;
    }
    public static String transform(String value) {
        return Transformer.valueOf(
            value.toUpperCase().equals(value) ? "UPPER" : "LOWER").operator.apply(value);
    }
}
```

B.
```
import java.util.function.UnaryOperator;
public enum Transformer {
    UPPER((String value) -> value.toUpperCase()),
    LOWER((String value) -> value.toLowerCase());
    private final UnaryOperator<String> operator;
    Transformer(UnaryOperator<String> operator) {
        this.operator = operator;
    }
    public static String transform(String value) {
        return Transformer.valueOf(
            value.toUpperCase().equals(value) ? "UPPER" : "LOWER").operator.apply(value);
    }
}
```

C.

```
import java.util.function.UnaryOperator;
public enum Transformer {
    UPPER((String value) -> value.toUpperCase()),
    LOWER((String value) -> value.toLowerCase());
    private final UnaryOperator<String> operator;
    public Transformer(UnaryOperator<String> operator) {
        this.operator = operator;
    }
    public static String transform(String value) {
        return switch(value) {
            case String v when v.toUpperCase().equals(value) -> LOWER.operator.apply(value);
            case String v when v.toLowerCase().equals(value) -> UPPER.operator.apply(value);
            default -> value;
        };
    }
}
```

D.

```
import java.util.function.UnaryOperator;
public enum Transformer {
    UPPER((String value) -> value.toUpperCase()),
    LOWER((String value) -> value.toLowerCase());
    private final UnaryOperator<String> operator;
    Transformer(UnaryOperator<String> operator) {
        this.operator = operator;
    }
    public static String transform(String value) {
        return switch(value) {
            case String v when v.toUpperCase().equals(value) -> LOWER.operator.apply(value);
            case String v when v.toLowerCase().equals(value) -> UPPER.operator.apply(value);
            default -> value;
        };
    }
}
```

E.

```
import java.util.function.UnaryOperator;
public enum Transformer {
    UPPER((String value) -> value.toUpperCase()),
    LOWER((String value) -> value.toLowerCase());
    private final UnaryOperator<String> operator;
    public Transformer(UnaryOperator<String> operator) {
        this.operator = operator;
    }
    public static String transform(String value) {
        return Transformer.valueOf(
            value.toUpperCase().equals(value) ? "U" : "L").operator.apply(value);
    }
    public String toString() {
        return switch(this) {
            case UPPER -> "U";
            case LOWER -> "L";
        };
    }
}
```

F.

```
import java.util.function.UnaryOperator;
public enum Transformer {
    UPPER((String value) -> value.toUpperCase()),
    LOWER((String value) -> value.toLowerCase());
    private final UnaryOperator<String> operator;
    Transformer(UnaryOperator<String> operator) {
        this.operator = operator;
    }
    public static String transform(String value) {
        return Transformer.valueOf(
            value.toUpperCase().equals(value) ? "U" : "L").operator.apply(value);
    }
    public String toString() {
        return switch(this) {
            case UPPER -> "U";
            case LOWER -> "L";
        };
    }
}
```

答案：D

第 38 題 第1次模擬考第 38 題

Given the `Listing` enum:

```
import java.util.*;
public enum Listing {
    LIST(new ArrayList<>()),
    SET(new HashSet<>());
    private Collection<String> values;
    Listing(Collection<String> values){
        this.values = values;
    }
    public Collection<String> setValues(String... values){
        for (String value: values) {
            this.values.add(value);
        }
        return this.values;
    }
}

// and the following code fragment:
String[] letters = {"A","B","B","A"};
StringBuilder txt = new StringBuilder();
for(int i = 0; i < Listing.values().length; i++) {
    Iterator<String> iter = Listing.values()[i].setValues(letters).iterator();
    while(iter.hasNext()) {
        String value = iter.next();
        txt.append(value);
    }
    System.out.print(txt);
}
```

What is the result?

A. `ABBA`

B. `ABBAAB`

C. `AB`

D. `ABB`

E. ABBAABBAAB

F. ABBAABBA

答案：E

6. 樣式比對與 switch

4 題

第 35 題

第1次模擬考第 35 題、第2次第 10 題、第3次第 26 題

Given:

```
class A {public void mA() {System.out.println("mA");}}
class B extends A {public void mB() {System.out.println("mB");}}
class C extends B {public void mC() {System.out.println("mC");}}

public class App {
    public static void main(String[] args) {
        A bobj = new B();
        A cobj = new C();
        if (cobj instanceof B v) {
            v.mB();
            if (v instanceof C v1) {
                v1.mC();
            }} else {
            cobj.mA();
        }
    }
}
```

What is the result?

A.
```
mB
```

B.
```
mB
mC
```

C.
```
mA
```

D.
```
mB
mA
```

答案：B

第 36 題

第1次模擬考第 36 題、第2次第 47 題、第3次第 42 題、第4次第 20 題

Given two record classes:

```
public record URL(Address<String, Integer> address, String value) { }
public record Address<V,P>(V host , P port) { }

// and the class:
public class Show {
    public static void main(String[] args) {
        URL url = new URL(new Address<>("localhost", Integer.valueOf(8080)),"hello");
        // Line n1
    }
}
```

Which code fragment prints `http://localhost:8080/hello` when inserted at Line n1?

A.
```
if (url instanceof URL(Address(var value, var port), var text) || !(port < 1024)) {
    System.out.println("http://"+value+":"+port+"/"+text);
}
```

B.
```
if (url instanceof URL(Address<>(var value, var port), var text) && port > 1024) {
    System.out.println("http://"+value+":"+port+"/"+text);
}
```

C.
```
if (url instanceof URL(Address<String,Integer> address, var text) && address.port > 1024) {
    System.out.println("http://"+address.value+":"+address.port+"/"+text);
}
```

D.
```
if (url instanceof URL(Address(var value, Integer port), var text) && port > 1024) {
    System.out.println("http://"+value+":"+port+"/"+text);
}
```

答案：D

第 37 題

第1次模擬考第 37 題、第3次第 44 題

Method `test` should accept any Object as an argument and return a String.

- If the argument is null or not an instance of `LocalDate`, it should return "Wrong Value".
- If the argument represents a date in the future, it should return "Future".
- If the argument represents a date in the past, it should return "Past".
- If the argument represents current date, it should return "Present".

Which code fragment correctly implements this scenario?

A.
```
public String test(Object obj) {
    return switch(obj) {
        default -> "Wrong Value";
        case null -> "Wrong Value";
        case LocalDate date when date.isBefore(LocalDate.now()) -> "Past";
        case LocalDate date when date.isAfter(LocalDate.now()) -> "Future";
        case LocalDate date when date.equals(LocalDate.now()) -> "Present";
    };
}
```

B.
```
public String test(Object obj) {
    return switch(obj) {
        case LocalDate date when date.isBefore(LocalDate.now()) -> "Past";
        case LocalDate date when date.isAfter(LocalDate.now()) -> "Future";
        case LocalDate date when !date.isBefore(LocalDate.now())
                                && !date.isAfter(LocalDate.now()) -> "Present";
        case !(LocalDate date) -> "Wrong Value";
        case null -> "Wrong Value";
    };
}
```

C.
```
public String test(Object obj) {
    return switch(obj) {
        case null -> "Wrong Value";
        case LocalDate date when date.isBefore(LocalDate.now()) -> "Past";
        case LocalDate date when date.isAfter(LocalDate.now()) -> "Future";
        case LocalDate date when date.equals(LocalDate.now()) -> "Present";
        case Object o -> "Wrong Value";
    };
}
```

D.

```
public String test(Object obj) {
    return switch(obj) {
        case LocalDate date when date.isBefore(LocalDate.now()) -> "Past";
        case LocalDate date when !date.isBefore(LocalDate.now()) -> "Future";
        case LocalDate date when date.equals(LocalDate.now()) -> "Present";
        default -> "Wrong Value";
    };
}
```

答案：C

第 39 題

第1次模擬考第 39 題、第2次第 49 題

Given:

```
public sealed interface Consumable permits Food, Drink { }
public record Food(String name) implements Consumable { }
public record Drink(String name) implements Consumable { }

// and a method:
public String test(Consumable consumable) {
    String result = "None";
    if (consumable instanceof Food(var name)) {
        result = "Eat "+name;
    } else if (consumable instanceof Drink(var name)) {
        result = "Drink "+name;
    }
    return result;
}
```

Which alternative code fragment implements behaviors that are equivalent to the given method example?

A.

```
public String test(Consumable consumable) {
    return switch (consumable) {
        case Food(String name) -> "Eat "+name;
        case Drink(String name) -> "Drink "+name;
        default -> "None";
    };
}
```

B.

```
public String test(Consumable consumable) {
    return switch (consumable) {
        default -> "None";
        case Food(String name) -> "Eat "+name;
        case Drink(String name) -> "Drink "+name;
    };
}
```

C.

```
public String test(Consumable consumable) {
    return switch (consumable) {
        case Food(String name) -> "Eat "+name;
        case default -> "Drink "+name;
        case null -> "None";
    };
}
```

D.

```
public String test(Consumable consumable) {
    String result = "None";
    switch (consumable) {
        case Food(String name): {
            result = "Eat "+name;
            break;
        }
        case Drink(String name): {
            result = "Drink " + name;
        }
    }
    return result;
}
```

E.

```
public String test(Consumable consumable) {
    return switch (consumable) {
        case null -> "None";
        case Food(String name) -> "Eat "+name;
        case Drink(String name) -> "Drink "+name;
    };
}
```

F.

```
public String test(Consumable consumable) {
    String result = switch (consumable) {
    case Food(String name): {
        yield "Eat " + name;
    }
    case Drink(String name): {
        yield "Drink " + name;
    }
    case null -> "None";
    };
    return result;
}
```

答案：E

7. 例外處理

3 題

第 14 題

第1次模擬考第 14 題、第2次第 46 題、第3次第 31 題

Given:

```
public class ContentException extends RuntimeException {
    public ContentException(String message) {
        super(message);
    }
}

public class ClosureException extends Exception {
    public ClosureException(String message) {
        super(message);
    }
}

public class TinCan implements AutoCloseable {
    private boolean problem;
    public TinCan(boolean problem) {
        this.problem = problem;
        System.out.println("opened");
    }
    public void checkContent(boolean anyGood) {
```

```
        if (!anyGood) {
            throw new ContentException("Not good");
        }
    }
    @Override
    public void close() throws ClosureException {
        if(problem) {
            throw new ClosureException("Problem closing the tin can");
        }
        System.out.println("closed");
    }
    public static void main(String[] args) {
        try (TinCan tc = new TinCan(true)) {
            tc.checkContent(false);
        } catch (ContentException | ClosureException e) {
            System.out.println(e);
        }
    }
}
```

Which two statements are true about this example?

(選 2 個)

A. The program will behave the same if the `catch` clause is replaced with `catch (Exception e)`.

B. It prints:

```
opened
ContentException: Not good
Problem closing the tin can
```

C. A `no-arg` constructor must be added to the `ClosureException`.

D. It prints:

```
opened
ContentException: Not good
Problem closing the tin can
closed
```

E. It prints:

```
opened
ContentException: Not good
```

F. A `no-arg` constructor must be added to the `ContentException`.

G. It prints:

```
opened
ContentException: Not good
closed
```

H. A `throws ContentException` clause must be added to the `checkContent` method.

答案：A, E

第 15 題

第1次模擬考第 15 題、第3次第 32 題

Given:

```
public class Test{
    public static void main(String[] args) {
        class ZeroValueException extends RuntimeException {
            ZeroValueException(String message) {
                super(message);
            }
            ZeroValueException(String message, Throwable cause) {
                super(message, cause);
```

```
            }
        }
        int nums[] = {1,0,1};
        int sum = 0;
        try {
            try {
                for (int v: nums) {
                    if (v != 0) {
                        sum += v;
                    }else{
                        throw new ZeroValueException("Zero value");
                    }
                }
            }catch(Exception e) {
                throw new ZeroValueException("Error",e);
            }
        }catch(Exception e) {
            System.out.print("Caught " + e.getMessage());
        }
        System.out.println(sum);
    }
}
```

What is the result?

A. `Caught Error1`

B. `Caught Zero value1`

C. `Caught Error2`

D. `Caught Zero value2`

答案：A

第 65 題

第2次模擬考第 45 題

Given:

```
import java.io.*;
import java.nio.file.*;
public class Test {
    public static void main(String[] args) {
        Test t = new Test();
        t.read("acme.txt");
    }
    public void read(String fileName) {
        String data = null;
        try {
            Path path = Path.of(fileName);
            openFile(path);
            readData(path);
        }catch(NoSuchFileException e){
            System.out.println(e.getMessage());
        }catch(Exception e){
            System.out.println("Something went wrong");
            return;
        }finally{
            System.out.println("Cleanup");
        }
        System.out.println("Data acquired "+data);
    }
    public void openFile(Path path) throws IOException {
        if (!Files.exists(path)){
```

```
            throw new NoSuchFileException("File not found");
        }
    }
    public String readData(Path path) throws IOException {
        return Files.lines(path)
                    .collect(StringBuilder::new,
                             StringBuilder::append,
                             StringBuilder::append)
                    .toString();
    }
}
```

Assume the following:

- The "acme.txt" file exists.
- It contains two lines of text:

  ```
  ONE
  TWO
  ```

- The storage for this file is corrupt and the reading of the second line fails.

What is the result?

A.
```
Something went wrong
```

B.
```
File not found
Cleanup
Data acquired null
```

C.
```
Something went wrong
Cleanup
Data acquired null
```

D.
```
Something went wrong
Cleanup
```

E.
```
Data acquired ONE
Something went wrong
Cleanup
```

F.
```
Something went wrong
Cleanup
Data acquired ONE
```

G.
```
Data acquired ONE
Something went wrong
```

答案：D

8. 集合與陣列

7 題

第 6 題

第1次模擬考第 6 題、第2次第 12 題、第3次第 49 題

Given:

```
Integer[] values = {-4,-2,12,7};
Arrays.sort(values, (v1, v2) -> v1.toString().compareTo(v2.toString()) );
System.out.println(Arrays.toString(values));
```

What is the result?

A.
```
[7, 12, -4, -2]
```

B.
```
[12, 7, -4, -2]
```

C.
```
[-2, -4, 12, 7]
```

D.
```
[-4, -2, 7, 12]
```

E.
```
[-4, -2, 12, 7]
```

F.
```
[7, 12, -2, -4]
```

G.
```
[12, 7, -2, -4]
```

答案：C

第 7 題 第1次模擬考第 7 題

Given:

```
List<String> list1 = new ArrayList<>();
list1.add("A");
list1.add("B");
List<String> list2 = Collections.unmodifiableList(list1);
list1.add("C");   // Line n1
System.out.println(list1);
System.out.println(list2); // Line n2
```

What is the result?

A.
```
[A, B, C]
[A, B, C]
```

B. On the line marked "Line n1", an exception is thrown at run time.

C.
```
[A, B, C]
[A, B]
```

D. `[A, B, C]`
followed by an exception thrown at line marked "Line n2".

答案：A

第 8 題 第1次模擬考第 8 題、第3次第 48 題

Which code fragment prints `Milk Honey Chocolate`?

A.
```
Set<String> foods = new HashSet<>(Arrays.asList("Milk","Honey","Chocolate"));
foods.forEach(f->System.out.print(f+" "));
```

B.
```
List<String> foods = new ArrayList<>(Arrays.asList("Chocolate","Milk","Honey"));
Collections.sort(foods, (f1,f2) -> f2.length()-f1.length());
foods.forEach(f->System.out.print(f+" "));
```

C.
```
List<String> foods = new ArrayList<>(Arrays.asList("Chocolate","Milk","Honey"));
foods.sort((f1,f2) -> f1.length()-f2.length());
foods.forEach(f->System.out.print(f+" "));
```

D.
```
Set<String> foods = new HashSet<>(Arrays.asList("Chocolate","Milk","Honey"));
Collections.sort(foods.stream().toList(),(f1,f2) -> f1.length()-f2.length());
foods.forEach(f->System.out.print(f+" "));
```

E.
```
Set<String> foods = new HashSet<>(Arrays.asList("Chocolate","Milk","Honey"));
Collections.sort(new ArrayList<String>(foods),(f1,f2) -> f1.length()-f2.length());
foods.forEach(f->System.out.print(f+" "));
```

答案：C

第 9 題 第1次模擬考第 9 題、第2次第 13 題、第3次第 46 題

Given:

```
String[] pets = {"Fido","Rex","Felix","Boo"};
```

Which two code fragments print the following text?

```
Fido
Fudge
Felix
Boo
```

（選 2 個）

A.
```
Set<String> petsSet = Set.of(pets);
petsSet.remove("Rex");
petsSet.add("Fudge");
petsSet.stream().forEach(p->System.out.println(p));
```

B.
```
List<String> petsList = Arrays.asList(pets);
pets[petsList.indexOf("Rex")] = "Fudge";
petsList.stream().forEach(p->System.out.println(p));
```

C.
```
Set<String> petsSet = new HashSet(Arrays.asList(pets));
petsSet.remove("Rex");
petsSet.add("Fudge");
petsSet.stream().forEach(p->System.out.println(p));
```

D.
```
List<String> petsList = Arrays.asList(pets);
petsList.removeIf(p->p.equals("Rex"));
petsList.add(1,"Fudge");
petsList.stream().forEach(p->System.out.println(p));
```

E.
```
List<String> petsList = Arrays.asList(pets);
petsList.stream()
        .map(p-> p.equals("Rex") ? "Fudge" : p)
        .forEach(p->System.out.println(p));
```

答案：B, E

第 53 題 第2次模擬考第 11 題

Given the code fragment:

```
String[] primes = {"2","3","5","7","11"};
Deque<Integer> numbers = new ArrayDeque<>();
for (int i = 0; i < primes.length; i++) {
    if (i != ((numbers.peekLast() != null) ? numbers.peekLast() : 0)) {
        numbers.offerFirst(Integer.parseInt(primes[i]));
    }else{
        numbers.offerLast(Integer.parseInt(primes[i]));
    }
}
for (Integer number : numbers) {
    System.out.print(number);
}
```

What is the result?

A. 117325

B. 117532

C. 253117

D. `273511`

E. `235711`

F. `112735`

答案：A

第 54 題 第2次模擬考第 14 題

Given the code fragment:

```
Map<String, BigDecimal> priceList = new HashMap<>();
priceList.put("Cake",BigDecimal.valueOf(3.99));
priceList.put("Cookie",BigDecimal.valueOf(1.99));
priceList.put("Candy",BigDecimal.valueOf(2.99));
priceList.put("Cookie",BigDecimal.valueOf(4.99));
priceList.entrySet().stream()
        .sorted((e1, e2)->e1.getValue().compareTo(e2.getValue()))
        .forEach(e->System.out.print(e+" "));
```

What is the result?

A. `Candy=2.99 Cake=3.99 Cookie=1.99`

B. `Cookie=1.99 Cake=3.99 Candy=2.99`

C. `Cookie=4.99 Cake=3.99 Candy=2.99`

D. Exception is thrown at run time.

E. `Cake=3.99 Cookie=1.99 Candy=2.99`

F. `Candy=2.99 Cake=3.99 Cookie=4.99`

G. `Cake=3.99 Cookie=4.99 Candy=2.99`

答案：F

第 73 題 第3次模擬考第 47 題

Which code fragment prints `Felix Boo Fido Rex`?

A.
```
SequencedCollection<String> pets = new TreeSet<>();
pets.addFirst("Rex");
pets.addFirst("Felix");
pets.addFirst("Fido");
pets.addFirst("Boo");
pets.addFirst("Felix");
for (String pet : pets) {
  System.out.print(pet+" ");
}
```

B.
```
SequencedCollection<String> pets = new LinkedHashSet<>();
pets.addFirst("Rex");
pets.addFirst("Felix");
pets.addFirst("Fido");
pets.addFirst("Boo");
pets.addFirst("Felix");
for (String pet : pets) {
  System.out.print(pet+" ");
}
```

C.
```
SequencedCollection<String> pets = new TreeSet<>();
pets.add("Rex");
pets.add("Felix");
pets.add("Fido");
pets.add("Boo");
pets.add("Felix");
for (String pet : pets) {
  System.out.print(pet+" ");
}
```

D.
```
SequencedCollection<String> pets = new LinkedHashSet<>();
pets.addFirst("Rex");
pets.addFirst("Felix");
pets.addFirst("Fido");
pets.addFirst("Boo");
for (String pet : pets) {
  System.out.print(pet+" ");
}
```

答案：B

9. Stream API

10 題

第 4 題

第1次模擬考第 4 題、第2次第 33 題、第4次第 50 題

Given the code fragment:

```
List<Integer> l1 = List.of(1, 3, 5, 7, 9);
List<Integer> l2 = List.of(2, 4, 6, 8);
List<List<Integer>> l3 = List.of(l1, l2);
//line n1
System.out.println(l4);
```

Which code fragment at `line n1` enables `l4` to contain numbers greater than 5 from the lists `l1` and `l2`?

A.
```
List<Integer> l4 = l3.stream()
        .map(x -> x.stream())
        .filter(y -> y > 5)
        .collect(Collectors.toList());
```

B.
```
List<Integer> l4 = l3.stream()
        .flatMap(x -> x.stream()
        .filter(y -> y > 5).toList());
```

C.
```
List<Integer> l4 = l3.stream()
        .filter(y -> y > 5).collect();
```

D.
```
List<Integer> l4 = l3.stream()
        .flatMap(x -> x.stream())
        .filter(y -> y > 5)
        .collect(Collectors.toList());
```

答案：D

第 5 題

第1次模擬考第 5 題

Given:

```
DoubleStream values = DoubleStream.generate(()-> ThreadLocalRandom.current().nextDouble());
double value = values.sorted().filter(d->d > 0.4 && d < 0.5)
                    .peek(d-> d++).findFirst().getAsDouble();
System.out.println(value);
```

What is the result?

A. The biggest double value between 0.4 and 0.5

B. The biggest double value between 1.4 and 1.5

C. An error is thrown at run time.

D. The smallest double value between 1.4 and 1.5

E. The smallest double value between 0.4 and 0.5

答案：C

第 48 題

第1次模擬考第 48 題

Given the code fragment:

```
Logger logger = Logger.getLogger("Test");
Stream<String> s1 = Stream.of("A", "B", "C", "B");
Stream<String> s2 = Stream.of("A", "D", "E");
Stream.concat(s1,s2).parallel().distinct().sorted()
        .forEach(e->logger.log(Level.INFO,e));
```

What is the result?

A. `ADEABCB`

B. `ABCDE`

C. `ABBCDE` // the order of elements is unpredictable

D. `ABBCDE`

E. `ADEABCB` // the order of elements is unpredictable

F. `ABCDE` // the order of elements is unpredictable

答案：F

第 49 題

第1次模擬考第 49 題、第2次第 25 題、第3次第 13 題

Given:

```
Stream<String> melody1 = Stream.of("E","G","A");
Stream<String> melody2 = Stream.of("D","C","F");
var value = Stream.concat(melody1, melody2).reduce((a,b)->b+a).get();
System.out.println(value);
```

What is the result?

A. `DCFEGA`

B. `AGEFCD`

C. `FCDAGE`

D. `EGADCF`

答案：C

第 50 題

第1次模擬考第 50 題、第2次第 27 題、第3次第 11 題

Given:

```
var list = List.of("Table", "Chair");
var result = list.parallelStream()
                 .reduce(list.parallelStream()
                             .reduce((l, p)-> l+p)
                             .get(),
```

```
                        (n,m) -> n+m, String::concat);
System.out.print(result);
```

What is the result?

A. `TableChairTableChair`

B. `TableChairTableChairTableChair`

C. `TableChairTableTableChairChair`

D. An exception is thrown at run time.

E. `TableChairTableChairChair`

答案：C

第 57 題 第2次模擬考第 26 題、第3次第 12 題

Given:

```
List<String> elements =
    Arrays.asList("car", "truck", "car",
                  "bicycle", "car", "truck", "motorcycle");
Map<String, Long> map =
    elements.stream()
            .collect(Collectors.groupingBy(Function.identity(), Collectors.counting()));
map.keySet().stream().sorted().forEach(k->System.out.print(map.get(k)));
```

What is the result?

A. `123`

B. `1312`

C. `1213`

D. `213`

E. `3211`

F. `2131`

G. `1123`

H. `312`

I. `321`

J. `7777`

答案：B

第 58 題 第2次模擬考第 28 題

Given the code fragment:

```
List<Integer> listOfNumbers = List.of(1, 2, 3, 4, 5, 6, 7, 8, 9, 10);
```

Which code fragment produces a `sum` value that is different from the other code fragments?

A. `int sum = listOfNumbers.parallelStream().reduce(5, Integer::sum);`

B. `int sum = listOfNumbers.stream().reduce(0, Integer::sum) + 5;`

C. `int sum = listOfNumbers.parallelStream().reduce(0, Integer::sum) + 5 ;`

D. `int sum = listOfNumbers.stream().reduce(5, (a, b) -> a+ b);`

E.
```
int sum = listOfNumbers.parallelStream().reduce((m, n) -> m + n).orElse(5) + 5;
```

答案：A

第 60 題 第2次模擬考第 32 題、第3次第 9 題

Given the code fragment:

```
Stream.of(3,6,9,12,15,18,21,24,27)
        .takeWhile(s -> s % 3 == 0)
        .dropWhile(s -> s % 2 == 0)
        .limit(3).forEach(s -> System.out.print(s + " "));
```

What is the result?

A.
```
3 9 15
```

B.
```
3 9 15 21
```

C.
```
3 6 9
```

D.
```
15 21 27
```

答案：C

第 70 題 第3次模擬考第 10 題

Given:

```
class Product {
    String name; double price;
    Product(String s, double d) {
        this.name = s;
        this.price = d;
    }
}
class ElectricProduct extends Product {
    ElectricProduct(String name, double price) {
        super(name, price);
    }
}

// and the code fragment:

List<Product> p = List.of(
    new ElectricProduct("CellPhone",100),
    new ElectricProduct("ToyCar",90),
    new ElectricProduct("Motor",200),
    new ElectricProduct("Fan",300)
);

DoubleSummaryStatistics sts = p.stream()
        .filter(a -> a instanceof ElectricProduct)
        .collect(Collectors.summarizingDouble(a -> a.price));
String s1 = p.stream().filter(a -> a instanceof Product)
        .collect(Collectors.mapping(p2 -> p2.name, Collectors.joining(",")));
System.out.println(sts.getMax());
System.out.println(s1);
```

What is the result?

A.
```
100.00 CellPhone,ToyCar,Motor,Fan
```

B.
```
100.00
CellPhone,ToyCar
```

C.
```
300.00
CellPhone,ToyCar,Motor,Fan
```

D.
```
300.00
CellPhone,ToyCar
```

答案：C

第 74 題 第3次模擬考第 50 題

Given:

```
public class TextGenerator {
  static int counter = 0;
  static int leftLimit = 'a';
  static int rightLimit = 'z';
  public static void main(String[] args) {
    // Line n1
  }
}
```

Which code fragment, when inserted into the main method at "Line n1", generates a text comprising random characters of 10000 characters in length?

A.
```
StringBuilder result = new StringBuilder();
ThreadLocalRandom.current()
  .ints(leftLimit,rightLimit+1)
  .limit(10000)
  .parallel()
  .forEach(c->result.append((char)c));
```

B.
```
String result = ThreadLocalRandom.current()
    .ints(leftLimit,rightLimit+1)
    .limit(10000)
    .parallel()
    .collect(StringBuilder::new, StringBuilder::appendCodePoint, StringBuilder::append)
    .toString();
```

C.
```
String result = ThreadLocalRandom.current()
    .ints(leftLimit,rightLimit+1)
    .takeWhile(c->counter++ < 10000)
    .parallel()
    .mapToObj(c-> String.valueOf((char)c))
    .collect(Collectors.joining());
```

D.
```
StringBuilder result = new StringBuilder();
ThreadLocalRandom.current()
  .ints(leftLimit,rightLimit+1)
  .takeWhile(c->result.length() != 10000)
  .parallel()
  .forEach(c->result.append((char)c));
```

答案：B

10. 並行與執行緒

8 題

第 10 題 第1次模擬考第 10 題

Which statements is true regarding how to achieve a thread-safe iteration through a collection?

A. Use the synchronized iterator provided by copy-on-write collections.

B. Implement your own synchronized iterator method to work with synchronized collections.

C. Use the synchronized iterator provided by immutable collections.

D. Implement your own synchronized iterator method to work with copy-on-write collections.

E. Implement your own synchronized iterator method to work with immutable collections.

F. Use the synchronized iterator provided by synchronized collections.

答案：A

第 11 題

第1次模擬考第 11 題、第2次第 24 題

Which statement is true?

A. The `lock()` method returns a boolean indicator immediately if it has managed to acquire the lock; otherwise, it waits for the lock acquisition.

B. The `lock()` method returns a boolean indicator immediately regardless of whether it has or has not managed to acquire the lock.

C. The `tryLock()` method returns a boolean indicator immediately if it has managed to acquire the lock; otherwise, it waits for the lock acquisition.

D. The `tryLock()` method returns a boolean indicator immediately regardless of whether it has or has not managed to acquire the lock.

答案：D

第 17 題

第1次模擬考第 17 題、第2次第 18 題

Which two statements are true about virtual threads?

(選 2 個)

A. Virtual threads can handle both `Callable` and `Runnable` objects.

B. The `Thread.setDaemon()` method is used to configure a virtual thread as daemon.

C. Virtual threads do not run code on the Platform threads.

D. Virtual threads are instances of the `java.lang.Thread` class.

E. The `Thread.setPriority()` method is automatically triggered to lower the priority of the Virtual thread when it is waiting for the I/O operations to complete.

F. The `Executors.newWorkStealingPool()` method is used to improve virtual thread performance by reusing threads in a thread pool.

答案：A, D

第 18 題

第1次模擬考第 18 題、第3次第 20 題

Which statement is true about thread life cycle?

A. The `Thread.interrupt()` method puts a thread into a timed waiting state.

B. The `Thread.interrupt()` method puts a thread into a waiting state.

C. Thread.interrupt() method puts a thread into a blocked state.

D. The `Thread.interrupt()` method does not change the state of a thread.

E. The `Thread.interrupt()` method puts a thread into a terminated state.

F. The `Thread.interrupt()` method puts a thread into a runnable state.

答案：D

第 19 題

第1次模擬考第 19 題、第2次第 19 題、第3次第 18 題

Which statement is true?

A. A thread in waiting state must handle `InterruptedException`.

B. After the timed wait expires, the waited thread moves to the terminated state.

C. `IllegalStateException` is thrown if a thread in waiting state is moved back to runnable.

D. A thread in waiting state consumes CPU cycles.

答案：A

第 55 題 第2次模擬考第 17 題、第3次第 19 題

Given the code fragment:

```
StringBuilder txt = new StringBuilder();
Runnable r = () -> {
    for (int i = 0; i < 5; i++) {
        synchronized(txt){
            txt.append(i);
        }
    }
};
Thread t1 = new Thread(r,"T1");
Thread t2 = new Thread(r,"T2");
ExecutorService es = Executors.newFixedThreadPool(2);
Future f1 = es.submit(t1);
Future f2 = es.submit(t2);
es.shutdown();
try {
    f1.get();
    f2.get();
    System.out.println(txt);
} catch (Exception e) {
    System.out.println("error");
}
```

What is the result?

A. Nothing is printed.

B. All integers from 0 to 4 are printed.

C. The word "error" is printed.

D. Some of the integers in the range 0 to 4 are printed.

E. Some of the integers in the range 0 to 4 are printed twice in stochastic order.

F. All integers from 0 to 4 are printed twice.

G. Some of the integers in the range 0 to 4 are printed twice.

H. All integers from 0 to 4 are printed twice in stochastic order.

I. All integers from 0 to 4 are printed in stochastic order.

J. Some of the integers in the range 0 to 4 are printed in stochastic order.

答案：H

第 68 題 第3次模擬考第 1 題

Given the code fragment:

```
ReentrantReadWriteLock rwl = new ReentrantReadWriteLock();
Lock rl = rwl.readLock();
Lock wl = rwl.writeLock();
String result = null;
try {
    rl.lock();
    if (wl.tryLock()) {
        result = "locked";
    }
```

```
} catch(Exception e) {
    result = "broken";
} finally {
    rl.unlock();
    if (rwl.isWriteLocked()) {
        wl.unlock();
        result = "unlocked";
    }
}
System.out.println(result);
```

What is the result?

A. `locked`

B. `broken`

C. `null`

D. `key`

E. `unlocked`

答案：C

第 75 題

第4次模擬考第 14 題

Given:

```
import java.util.concurrent.atomic.AtomicInteger;
import java.util.logging.Level;
import java.util.logging.Logger;
class Test {
    volatile int x = 1;
    AtomicInteger y = new AtomicInteger(1);
    public static void main(String[] args) {
        Logger logger = Logger.getLogger("p6.Test");
        Test t = new Test();
        Runnable r1 = () -> {
            while (t.x < 3) {
                logger.log(Level.INFO,"Thread One :"+String.valueOf(t.x));
                t.x++;
            }
        };
        Runnable r2 = () -> {
            while (t.y.get() < 3) {
                logger.log(Level.INFO,"Thread Two :"+String.valueOf(t.y));
                t.y.getAndIncrement();
            }
        };
        Thread t1 = new Thread(r1);
        Thread t2 = new Thread(r2);
        t1.setPriority(4);
        t2.setPriority(7);
        t1.start();
        t2.start();
    }
}
```

Which statement is true?

A. The program logs values 1 and 2 for the first thread and then values 1 and 2 for the second thread.

B. The program logs values 1 and 2 for each of the thread in random order.

C. The program throws an exception.

D. The program logs values 1 and 2 for the second thread and then values 1 and 2 for the first thread.

E. The program logs values 1 and 2 for the second thread and then value 1 for the first thread indefinitely.

答案：B

11. I/O、NIO 與序列化

6 題

第 16 題

第1次模擬考第 16 題、第3次第 33 題

Given the code fragment:

```
Path p1 = Path.of("fldr1/fldr2/file1.txt");
Path p2 = Path.of("fldr3/file1.txt");
System.out.println(p1.resolve(p2));
System.out.println(p1.relativize(p2));
```

What is the result?

A.
```
fldr1/fldr2/fldr3/file1.txt
../../../../fldr3/file1.txt
```

B.
```
fldr1/fldr2/file1.txt/fldr3/file1.txt
fldr3/file1.txt
```

C.
```
fldr1/fldr2/file1.txt/fldr3
../../../fldr3/
```

D.
```
fldr1/fldr2/file1.txt/fldr3/file1.txt
../../../fldr3/file1.txt
```

答案：D

第 28 題

第1次模擬考第 28 題、第2次第 16 題、第3次第 27 題

Given the `Software` class:

```
import java.io.Serializable;
public class Software implements Serializable {
    private String title;
    public Software(String title) {
        this.title = title;
        System.out.print("Software ");
    }
    public String toString() { return title; }
}

// and the Game class:
public class Game extends Software {
    private int players;
    public Game(String title, int players) {
        super(title);
        this.players = players;
        System.out.print("Game ");
    }
    public String toString() { return super.toString()+" "+players; }
}

// and the AppStore class:
import java.io.*;
public class AppStore {
    public static void main(String[] args) {
        Software s = new Game("Chess", 2);
```

```
        try(ObjectOutputStream out = new ObjectOutputStream(new FileOutputStream("game.ser"))) {
            out.writeObject(s);
        } catch (Exception e) {
            System.out.println("write error");
        }
        try(ObjectInputStream in = new ObjectInputStream(new FileInputStream("game.ser"))) {
            s = (Software)in.readObject();
        } catch (Exception e) {
            System.out.println("read error");
        }
        System.out.println(s);
    }
}
```

What is the result?

A. `Software Game Chess 2`

B. `Software Game Software Game Chess 0`

C. `Software Game write error`

D. `Software Game Software Game Chess 2`

E. `Software Game Chess 0`

F. `Software Game read error`

答案：A

第 29 題

第1次模擬考第 29 題、第2次第 15 題、第3次第 28 題

Given the `Product` class:

```
import java.io.*;
public class Product implements Serializable {
    private static float averagePrice = 2.99f;
    private String description;
    private transient float price;
    public Product(String description, float price) {
        this.description = description;
        this.price = price;
    }
    public void readObject(ObjectInputStream in)
            throws IOException, ClassNotFoundException {
        in.defaultReadObject();
        price = averagePrice;
    }
    public String toString() {
        return description+" "+price+" "+averagePrice;
    }
}

// and the Shop class:
import java.io.*;
public class Shop {
    public static void main(String[] args) {
        Product p = new Product("Cookie", 3.99f);
        try {
            try (ObjectOutputStream out = new ObjectOutputStream(new FileOutputStream("p.ser")))
{
                out.writeObject(p);
            }
            try (ObjectInputStream in = new ObjectInputStream(new FileInputStream("p.ser"))) {
                p = (Product)in.readObject();
```

```
            }
        } catch (Exception e) { e.printStackTrace(); }
        System.out.println(p);
    }
}
```

What is the result?

A. `Cookie 2.99 2.99`

B. `Cookie 0.0 2.99`

C. `Cookie 3.99 2.99`

D. An exception is produced at run time.

E. Compilation fails.

F. `Cookie 0.0 0.0`

答案：B

第 32 題 第1次模擬考第 32 題、第2次第 4 題、第3次第 6 題

Which statement is true?

A. The `java.lang.System.console()` method throws an exception when console is not available.

B. The `java.io.PrintStream` class converts characters to bytes when printing to the console.

C. The `java.io.PrintWriter` class converts characters to bytes when printing to the console.

D. The `java.io.Console.readPassword()` method encrypts text typed by the user.

答案：B

第 33 題 第1次模擬考第 33 題、第2次第 3 題、第3次第 7 題

Given the content of the `in.txt` file:

```
racecar
```

and the code fragment:

```
char[] buffer = new char[4];
int count = 0;
try(FileReader in = new FileReader("in.txt");
    FileWriter out = new FileWriter("out.txt")) {
    while((count = in.read(buffer)) != -1) {
        out.write(buffer);
    }
}
```

What is the content of the `out.txt` file?

A. `racecara`

B. `racecare`

C. `raceca`

D. `race`

E. `racec`

F. `racecar`

答案：B

第 64 題 第2次模擬考第 44 題

Assuming that the `/test` folder contains two files:
`infoA.txt`, which contains these two lines of text:

```
a!
b!!
```

and `infoB.txt`, which contains these two lines of text:

```
c?!
?d?
```

and given this code fragment:

```
int result = Files.list(Path.of("/test"))
                .filter(p->Files.isRegularFile(p) &&
                        p.toString().endsWith("txt"))
                .flatMap(p -> {
                  try {
                    return Files.lines(p);
                  } catch (IOException e) {
                    return null; }})
                .filter(s->s.contains("!"))
                .map(s->s.indexOf("?"))
                .reduce((s1,s2) -> s1+s2).get();
System.out.println(result);
```

What is the result?

A.
```
2
```

B.
```
-1
```

C.
```
0
```

D.
```
-2
```

E.
```
1
```

答案：B

12. 日期時間、格式化與本地化

5 題

第 1 題 第1次模擬考第 1 題

Given the code fragment:

```
int saleAmount = 10000;
NumberFormat format1 = NumberFormat.getCurrencyInstance(Locale.US);
NumberFormat format2 = NumberFormat.getCompactNumberInstance(Locale.US, Style.SHORT);
String res = format1.format(saleAmount);
System.out.println(res);
format2.format(Integer.valueOf(res));
System.out.println(saleAmount);
```

What is the result?

A.
```
$10,000.00
$10K
```

B.
```
$10000.00
$10.00 K
```

C.
```
$10,000.00
$10,000.00 K
```

D. `$10,000.00`
An exception is thrown.

答案：D

第 46 題 第1次模擬考第 46 題、第2次第 8 題

Which code fragment prints `2024-05-04`?

A.
```
var x = LocalDate.of(2024, 1, 4)
                 .plusMonths(1)
                 .minus(Period.ofDays(-3));
System.out.println(x);
```

B.
```
var x = LocalDate.of(2024, 4, 1);
x.withDayOfMonth(4).plus(Period.ofDays(30));
System.out.println(x);
```

C.
```
var x = LocalDate.of(2024, 4, 1)
                 .plusMonths(1)
                 .withDayOfMonth(15)
                 .plusDays(-4)
                 .minus(Period.ofWeeks(1));
System.out.println(x);
```

D.
```
var x = Period.between(
          LocalDate.of(2024, 4, 1),
          LocalDate.of(2024, 5, 1).plusDays(4));
System.out.println(x);
```

答案：C

第 47 題 第1次模擬考第 47 題、第3次第 30 題

Given:

```
LocalDate d1 = LocalDate.of(2021, Month.JANUARY, 1);
LocalDate d2 = d1.plus(Period.ofDays(-10)).withMonth(1);
LocalDateTime ldt = d2.atTime(LocalTime.of(11, 11));
if (d1.isAfter(d2) ^! d1.getMonth().equals(ldt.getMonth())) {
    ldt = d2.atTime(LocalTime.of(12, 12));
}
System.out.println(ldt.format(DateTimeFormatter.ofPattern("HHMM")));
```

What is the result?

A. `1111`

B. `1201`

C. `1101`

D. A runtime exception is thrown.

E. `1212`

答案：B

第 52 題 第2次模擬考第 7 題、第3次第 29 題

Daylight Saving Time (DST) is the practice of advancing clocks at the start of spring by one hour and adjusting them backward by one hour in autumn.

Considering that in 2021, DST in Chicago (Illinois) ended on November 7th at 2 AM, and given the fragment:

```
ZoneId zoneID = ZoneId.of("America/Chicago");
ZonedDateTime zdt = ZonedDateTime.of(
    LocalDate.of(2021, 11, 7),
    LocalTime.of(1, 30),
    zoneID);
ZonedDateTime anHourLater = zdt.plusHours(1);
System.out.println(zdt.getHour() == anHourLater.getHour());
System.out.print(zdt.getOffset().equals(anHourLater.getOffset())
);
```

What is the output?

A.
```
false
true
```

B.
```
false
false
```

C.
```
true
false
```

D.
```
true
true
```

答案：C

第 59 題 第2次模擬考第 31 題、第3次第 8 題

Given:

```
Locale.setDefault(Locale.US);
Locale currentLocale = new Locale.Builder().setLanguage("en").build();

ResourceBundle captions = ResourceBundle.getBundle("Captions.properties", currentLocale);
System.out.println(captions.getString("user"));
```

```
Captions.properties file:
user = UserName

Captions_en.properties file:
user = User name (EN)

Captions_US.properties file:
message = User name (US)

Captions_en_US.properties file:
message = User name (EN - US)
```

and the code fragment:

What is the result?

A.
```
User name (EN - US)
```

B.
```
User name (US)
```

C.
```
User name (EN)
```

D. The program throws a MissingResourceException

E.
```
UserName
```

答案：D

13. 模組系統

5 題

第 2 題

第1次模擬考第 2 題

Given the code fragment:

```
module citizen {
    exports com.name to greeting;
}

module greeting {
}
```

Which statement is true?

A. Public members in the `citizen` module are accessible only to the `greeting` module.

B. Public members in the `com.name` package are accessible to all modules.

C. All members of `com.name` are accessible only to the `citizen` and `greeting` modules.

D. Inserting `requires citizen;` into the `greeting` module-info enables classes in the `greeting` module to access public members of the `com.name` package.

E. Inserting `opens com.name;` into the `greeting` module-info enables classes in the `greeting` module to access all members of the `com.name` package.

F. All members in the `com.name` package are accessible only to the `greeting` module.

答案：D

第 3 題

第1次模擬考第 3 題、第2次第 35 題、第3次第 16 題

Given the following scenario:

- A module `abc` contains an `abc.shopping.ShopService` interface.
- A module `xyz` contains an `xyz.shop.OnlineShop` class that implements the `abc.shopping.ShopService` interface.
- A module `acme` needs to use the implementation of `ShopService` provided by the `xyz` module.

Which statement is true?

A. A `requires xyz` directive must be added to the `acme` module-info.

B. An `opens abc` directive must be added to the `acme` module-info.

C. A `requires abc.shopping` directive must be added to the `acme` module-info.

D. A `requires xyz.shop` directive must be added to the `acme` module-info.

E. An `opens abc.shopping` directive must be added to the `acme` module-info.

F. An `opens xyz.shop` directive must be added to the `acme` module-info.

G. A `requires abc` directive must be added to the `acme` module-info.

H. An `opens xyz` directive must be added to the `acme` module-info.

答案：G

第 22 題

第1次模擬考第 22 題、第2次第 39 題、第3次第 38 題

Which two statements are true about running code on the class path and the module path?

（選 2 個）

A. A modular JAR placed on the `--module-path` results in a named application module.

B. A non-modular JAR placed on the `--module-path` results in a named application module.

C. A non-modular JAR placed on the `-classpath` results in an unnamed module.

D. A modular JAR placed on the `-classpath` results in an automatic module.

E. A modular JAR placed on the `-classpath` results in a named application module.

答案：A, C

第 23 題 第1次模擬考第 23 題、第2次第 38 題、第3次第 39 題

Which statement is true regarding the JImage deployment?

A. A JImage can only include Java artefacts such as compiled classes and property bundles.

B. A JImage can only include modules that are referred to via the requires, or opens directives.

C. A JImage can be produced to run on multiple platforms.

D. A JImage can be produced for a non-modular application.

答案：C

第 61 題 第2次模擬考第 34 題、第3次第 17 題

Given:

```
package com.transport.vehicle.cars;
public interface Car {
    int getSpeed();
}

// and

package com.transport.vehicle.cars.impl;
import com.transport.vehicle.cars.Car;
public class CarImpl implements Car {
    private int speed;
    public CarImpl() {
        this(10);
    }
    public CarImpl (int speed) {
        this.speed = speed;
    }
    @Override
    public int getSpeed() {
        return speed;
    }
}
```

Which two should the module-info file include for it to represent the service provider interface?

（選 2 個）

A. `provides com.transport.vehicle.cars.Car with com.transport.vehicle.cars.impl.CarImpl;`

B. `exports com.transport.vehicle.cars;`

C. `provides com.transport.vehicle.cars.impl.CarImpl for com.transport.vehicle.cars.Car;`

D. `exports com.transport.vehicle.cars.Car;`

E. `exports com.transport.vehicle;`

F. `requires com.transport.vehicle.cars.Car;`

G. `requires com.transport.vehicle.cars;`

答案：A, B